1. Core Concepts & Modelling

Dynamical system: state $x(t)$ evolves from current state + inputs/disturbances. **Feedback**: measure output, compare to reference r, adjust input u.
LTI state-space:

$$\dot{\mathbf{x}} = A\mathbf{x} + Bu + D, \quad y(t) = C\mathbf{x}$$

A=dynamics (main), B=input, C=output, D=disturbance. $A\in\mathbb{R}^{n\times n}$, $B\in\mathbb{R}^{n\times 1}$, $C\in\mathbb{R}^{1\times n}$. Feedthrough form: $y=C\mathbf{x}+Du$.
ODE → SS: for $\ddot{x}+a\dot{x}+bx=u$, $\mathbf{x}=(x,\dot{x})^T$:

$$A=\begin{pmatrix}0 & 1\\ -b & -a\end{pmatrix},\ B=\begin{pmatrix}0\\1\end{pmatrix},\ C=(1 \quad 0)$$

TF → SS: for $G(s)=\dfrac{b_0}{s^2+a_1s+a_0}$:

$$A=\begin{pmatrix}0 & 1\\ -a_0 & -a_1\end{pmatrix},\ B=\begin{pmatrix}0\\1\end{pmatrix},\ C=(b_0 \quad 0)$$

Mass-spring-damper: $m\ddot{y}+c\dot{y}+ky=u$, $G(s)=\dfrac{1}{ms^2+cs+k}$.

2. Linearisation (Lec 2)

For $\dot{x}=f(x,u)$:
(1) Equilibrium $f(x_e,u_e)=0$.
(2) Taylor (1st order):

$$\dot{x}\approx f(x_e,u_e)+\frac{\partial f}{\partial x}\Big|_e(x-x_e)+\frac{\partial f}{\partial u}\Big|_e(u-u_e)$$

(3) Deviation vars $\delta x=x-x_e$, $\delta u=u-u_e$:

$$\dot{\delta x}\approx\frac{\partial f}{\partial x}\Big|_e\delta x+\frac{\partial f}{\partial u}\Big|_e\delta u$$

Multivariable: $A=\frac{\partial f}{\partial \mathbf{x}}\big|_e$ (Jacobian), $B=\frac{\partial f}{\partial u}\big|_e$.

3. First-Order Systems (Lec 3)

$\dot{x}+ax=u$, free resp.: $x(t)=e^{-at}x(0)$.
Stability: pole at $s=-a$. $a>0\Rightarrow$ decays, **stable**; $a<0\Rightarrow$ grows, **unstable**.
Step (u=const), $x(0)=x_0$:

$$x(t)=\left(x_0-\frac{u}{a}\right)e^{-at}+\frac{u}{a}$$

Steady state: $x(\infty)=u/a$. Time constant $\tau=1/a$.
Time specs: rise $t_r\approx\frac{2.2}{a}=2.2\tau$ (10–90%); settle $t_s\approx\frac{3.91}{a}\approx\frac{4}{a}=4\tau$ (2%).
Transfer function: $G(s)=\dfrac{\text{output}}{\text{input}}=\dfrac{Y(s)}{U(s)}$, $y=G(s)u$, $u=Ue^{st}$.
$G(s)=\dfrac{X(s)}{U(s)}$. For $\ddot{x}+3\dot{x}+2x=u$: $G=\frac{1}{s^2+3s+2}$.
DC gain → final value: for *any stable* G, step of size U:

$$x(\infty)=G(0)\cdot U \quad \text{(set all derivs = 0).}$$

$G(0)$ sign = direction ($G(0)<0\Rightarrow$ output dips below 0); magnitude = final value. Generalises $x_\infty=u/a$.

4. Proportional / Steady-State Error

Plant $\dot{x}+ax=u$, $u=K_P(r-x)$:

$$\dot{x}+(a+K_P)x=K_Pr$$

$x(\infty)=\frac{K_Pr}{K_P+a}$, error $e(\infty)=r\,\frac{a}{K_P+a}$.
With disturbance d, noise n:

$$x_{ss}=\frac{K_P(r-n)+d}{a+K_P}$$

Large $K_P\Rightarrow$ small error (**feedback stabilisation**: pick $K_P>-a$). **Integral** action $\Rightarrow$ zero steady-state error.
SS error (unity fb, unit step): $e(\infty)=\dfrac{1}{1+G(0)}$ ($G(0)$=DC loop gain; position const K_p=$G(0)$).

5. Homogeneous Solns & Poles (Lec 4)

Char. eq. of $\ddot{x}+a\dot{x}+bx=0$: $\lambda^2+a\lambda+b=0$.
Real distinct: ($a^2-4b>0$): $x_h=C_1e^{\lambda_1t}+C_2e^{\lambda_2t}$.
Real repeated: ($a^2-4b=0$): $x_h=(C_1+C_2t)e^{\lambda^*t}$.
Complex: $\lambda=-\sigma\pm j\omega_d$:

$$x_h=e^{-\sigma t}(C_3\cos\omega_dt+C_4\sin\omega_dt)$$

Stability: all poles (roots of denom.) have negative real part $\Leftrightarrow x(t)\to0$ for any $x(0)$, zero input.
Necessary cond.: char. poly has *all coefficients present & positive* (same sign) — any missing/sign change $\Rightarrow$ unstable. Real, distinct, negative poles $\Rightarrow$ stable & non-oscillatory.
$G(s)=\dfrac{b_1s+b_0}{s^2+a_1s+a_0}$; **poles**=denom roots, **zeros**=numer roots.

6. Second-Order Systems (Lec 5)

Standard: $G(s)=\dfrac{\omega_n^2}{s^2+2\zeta\omega_ns+\omega_n^2}$.
Poles $s=-\zeta\omega_n\pm j\omega_n\sqrt{1-\zeta^2}=-\sigma\pm j\omega_d$.

$$\sigma=\zeta\omega_n,\quad \omega_d=\omega_n\sqrt{1-\zeta^2}$$

$$\omega_n=\sqrt{\sigma^2+\omega_d^2},\quad \zeta=\sigma/\omega_n$$

ζ = sine of angle from imag. axis. Underdamped: $0<\zeta<1$.
Envelope: $x(t)=e^{-\sigma t}\sin(\omega_dt)$.
Response specifications:

$$M_p=e^{-\pi\zeta/\sqrt{1-\zeta^2}} \quad \text{(overshoot)}$$

$$t_p=\frac{\pi}{\omega_d} \quad \text{(peak time)}$$

$$t_s\approx\frac{4}{\sigma}=\frac{4}{\zeta\omega_n} \quad \text{(2\% settling)}$$

$$t_r\approx\frac{1.8}{\omega_n} \quad \text{(rise 10–90\%)}$$

5% overshoot $\Rightarrow\zeta\approx\frac{\sqrt{2}}{2}\approx0.707$.

7. PID Control (Lec 5)

$$C(s)=K_p+\frac{K_i}{s}+K_ds=\frac{K_ds^2+K_ps+K_i}{s}$$

- **P**: fast, may leave SS error / oscillate.
- **PI**: removes SS error, slower transient.
- **PD**: adds damping, faster, noise-sensitive, adds a zero.
- **PID**: removes SS error + improves transient.

$K_p\uparrow\Rightarrow\omega_n\uparrow$ (faster), $\zeta\downarrow$ (more overshoot).
Rate feedback: $u=K_P(r-x)-K_D\dot{x}$ (avoids derivative kick):

$$G(s)=\frac{K_P+K_Ds}{s^2+(1+K_D)s+K_P}$$

zero at $s=-K_P/K_D$.
PD design ex. $\ddot{x}+3\dot{x}+2x=u$, $u=K_p(r-x)-K_d\dot{x}$:

$$G=\frac{K_p}{s^2+(3+K_d)s+(K_p+2)}$$

match to $s^2+2\zeta\omega_ns+\omega_n^2$.

8. Effect of Zeros (Lec 5)

$G(s)=\dfrac{b_1s+b_0}{s+a}$, zero at $s=-b_0/b_1$.
$b_1>0,b_0>0$: LHP zero, **min-phase**, kick same dir.
$b_1<0,b_0>0$: RHP zero, **non-min-phase**, "wrong-way" initial response, hard to control.

9. State-Space Response (Lec 6)

Matrix exponential: $e^{At}=\displaystyle\sum_{k=0}^{\infty}\frac{(At)^k}{k!}=I+At+\frac{(At)^2}{2!}+\cdots$
Free resp.: $\mathbf{x}(t)=e^{At}\mathbf{x}(0)$.
Forced: $\mathbf{x}(t)=e^{At}\mathbf{x}_0+\displaystyle\int_0^t e^{A(t-\tau)}Bu\,d\tau$.
TF from SS: $G(s)=C(sI-A)^{-1}B+D$.
Output to $u=Ue^{st}$: $y=Ce^{At}\mathbf{x}_0+G(s)Ue^{st}$.
Stable $\Leftrightarrow$ all eigenvalues of A have Re <0.
$e^{At}=Te^{Jt}T^{-1}$ (Jordan); repeated eig give t^k terms.

10. Reachability / Control (Lec 7)

Reachability matrix: (2nd order: 2×2)

$$W=[B \quad AB]$$

Reachable $\Leftrightarrow\det W\neq0$.
Equivalent: can place *any* closed-loop poles via K.
State feedback: $u=-K\mathbf{x}$, closed loop $\dot{\mathbf{x}}=(A-BK)\mathbf{x}$.
Design: $\det(sI-(A-BK))\overset{!}{=}$ desired char. poly, **match coefficients**.
Transformation: $T=W\tilde{W}^{-1}$ ($\tilde{W}$ from reachable canonical form), $K=\tilde{K}T$.
Reachable canonical form: for $\ddot{y}+a_1\dot{y}+a_2y=u$ (top-row companion):

$$\tilde{A}=\begin{pmatrix}-a_1 & -a_2\\ 1 & 0\end{pmatrix},\ \tilde{B}=\begin{pmatrix}1\\0\end{pmatrix}$$

char. poly $s^2+a_1s+a_2$ read off top row.
Dominant poles: choose 2nd-order pair for transient; place rest 5–10× further left, low overshoot.

11. Observability / Observers (Lec 8)

Observability matrix: (2nd order: 2×2)

$$W_o=\begin{bmatrix}C\\CA\end{bmatrix}$$

Observable $\Leftrightarrow\det W_o\neq0$: recover $\mathbf{x}(0)$ from u,y on $[0,T]$.
Luenberger observer:

$$\dot{\hat{\mathbf{x}}}=A\hat{\mathbf{x}}+Bu+L(y-C\hat{\mathbf{x}})$$

Error $\tilde{\mathbf{x}}=\mathbf{x}-\hat{\mathbf{x}}$: $\dot{\tilde{\mathbf{x}}}=(A-LC)\tilde{\mathbf{x}}$.
Condition: $A-LC$ stable ($\Rightarrow\tilde{\mathbf{x}}\to0$). Roots (eig) of $A-LC$: **negative** $\Rightarrow$ error decays, $\hat{\mathbf{x}}\to\mathbf{x}$ (convergent); **positive** $\Rightarrow$ unstable, error grows, never converges.
Design: $\det(sI-(A-LC))\overset{!}{=}$ desired poly, match coeffs.
Output feedback: (separation): observer + $u=-K\hat{\mathbf{x}}+Nr$. CL char poly: $\det(sI-(A-BK))\det(sI-(A-LC))$.
Separation principle: applies to systems that are *both reachable &*

observable: design K and L **independently**; the CL poles are just the union of the controller poles (eig $A-BK$) and observer poles (eig $A-LC$).
Feedforward: (zero SS err): $N = -\big(C(A-BK)^{-1}B\big)^{-1}$.
Integral action: augment state $\dot q = C\mathbf{x} - \bar y$ to kill SS error without exact model.

12. Frequency Response (Lec 9)

Input $A\sin\omega t \Rightarrow y = A|G(j\omega)|\sin(\omega t+\phi)$.
$G(j\omega) = |G(j\omega)|e^{j\phi}$, $\phi = \angle G = \arctan\frac{\mathrm{Im}}{\mathrm{Re}}$.
$\angle G = \sum\angle(\text{num}) - \sum\angle(\text{den})$: **numerator** (zeros) add +ve phase, **denominator** (poles) add −ve phase.
Complex algebra: z_1z_2: multiply mags, add phases; z_1/z_2: divide mags, subtract phases.
Ex. $G = \dfrac{1}{j\omega(j\omega+1)(j\omega+3)}$:

$$|G| = \frac{1}{\sqrt{\omega^2(\omega^2+1)(\omega^2+9)}}$$

$$\angle G = -\big(\tfrac{\pi}{2} + \arctan\omega + \arctan\tfrac{\omega}{3}\big)$$

Forced sinusoid (undetermined coeffs): for $\ddot x + \omega_n^2 x = \sin\omega t$, try $x{=}X\sin\omega t$, match:

$$X = \frac{1}{\omega_n^2 - \omega^2}$$

real $\Rightarrow$ phase $0^\circ/180^\circ$ (no lag); $X\to\infty$ as $\omega\to\omega_n$ (**resonance**, undamped).

13. Bode Plots (Lec 9–10)

Mag in dB: $20\log_{10}|G(j\omega)|$. $K = 10^{dB/20}$, $dB = 20\log_{10}K$. $\times10 \Rightarrow$ 20dB; $1\to$0dB.
Standard form: factor so each bracket starts with 1; constant out front = DC gain K:

$$G(s) = \frac{K(\frac{s}{a}+1)}{s^n(\frac{s}{b}+1)}$$

Start height (flat, low ω) = $20\log_{10}K$ dB. If sloped (i.e pole @ 0), choose small ω and S.H. = $20\log_{10}\left(\dfrac{K}{\omega}\right)$.
Mag slopes: (added at break freq $\omega{=}|p|,|z|$):

DC gain K	flat at $20\log K$
integrator $1/s^n$	$-20n$ dB/dec from $\omega{\to}0$
diff. s	+20 dB/dec
pole $(\frac{s}{b}{+}1)^{-1}$	−20 dB/dec after $\omega{=}b$
zero $(\frac{s}{a}{+}1)$	+20 dB/dec after $\omega{=}a$
cplx poles @ ω_n	−40 dB/dec; peak $\frac{1}{2\zeta}$ at ω_n
cplx zeros @ ω_n	+40 dB/dec

Phase: $\angle G = \sum\angle(\text{num}) - \sum\angle(\text{den})$; each $(j\omega{+}a)$: $\arctan\frac{\omega}{a}$; transition ±1 decade about corner.

DC gain	0°
integrator $1/s$	-90° (all ω)
diff. s	$+90^\circ$ (all ω)
pole @ b	-90° total, centred b
zero @ a	$+90^\circ$ total, centred a
cplx poles @ ω_n	-180° total

Bode → TF: **(1)** DC gain $K{=}10^{X\,\mathrm{dB}/20}$ from flat low-ω ($\omega = 1$ if slanted start). **(2)** initial slope: flat=no int., −20/dec=one $1/s$, $-40{=}1/s^2$, $+20{=}s$. **(3)** corners where slope *changes*: extra $-20 \Rightarrow$ pole, $+20\Rightarrow$ zero, $\mp40 \Rightarrow$ cplx pair @ ω_n. **(4)** phase confirms: start $0/{-90}/{-180}^\circ$=0/1/2 integrators; drop @ corner=pole, rise=zero. **(5)** assemble:

$$G(s) = \frac{K\,s^{\text{pos}}(s+z_1)(s+z_2)\cdots}{s^{\text{int}}(s+p_1)(s+p_2)\cdots}$$

Sanity: final slope = $-20(\#p-\#z)$ dB/dec; final phase = $-90^\circ(\#p-\#z)$; DC check: sub $s{=}0$.
GM & PM on Bode: **GM** = magnitude (dB) at phase = -180°; **PM** = phase (deg) at magnitude 0 dB.
Ex.: flat +20dB, corner $\omega{=}2$ ($\to -20$), corner $\omega{=}20$ ($\to -40$), phase $0\to-90\to-180^\circ$: $K{=}10$, poles $2, 20$:

$$G = \frac{10}{(\frac{s}{2}+1)(\frac{s}{20}+1)} = \frac{400}{(s+2)(s+20)}$$

14. Nyquist Criterion (Lec 10,12)

Plot $L(j\omega)$, $\omega: -\infty\to\infty$. Closed loop $G_{yr} = \frac{L}{1+L}$.
Theorem: P=open-loop RHP poles of L; N=net *clockwise* encirclements of −1. Then CL has $Z = N + P$ RHP poles.
Stable $\Leftrightarrow Z = 0 \Leftrightarrow N = -P$ (P anticlockwise encirclements).
Stable-L case: stable iff *no* encirclements of −1.
CL poles satisfy $L(s) = -1$.
Real-axis crossing: Nyquist evaluates G along the imag. axis, so to find where it crosses $(\alpha, 0)$: set $G(s){=}\alpha$ and solve — the root is **purely imaginary** $s{=}\pm j\omega$.

15. Stability Margins (Lec 10,11)

Crossover ω_{gc}: $|L(j\omega_{gc})| = 1$ (0 dB).
Gain margin: $g_m = 1/|L(j\omega_{pc})|$ at phase $= -180^\circ$. *Find where Nyquist crosses −ve real axis; GM = inverse of that magnitude. (+ve).*
Phase margin: $\varphi_m = 180^\circ + \angle L(j\omega_{gc})$. *Find where Nyquist hits unit circle; PM = angle from that point to −ve x-axis.*
Stability margin: s_m=closest dist of Nyquist to −1; $|S(j\omega)|$ = 1/dist to (−1). *Shortest distance from Nyquist curve to* $(-1, 0)$.
$PM = 2\sin^{-1}\big(\frac{1}{2|S(j\omega_c)|}\big)$.
Time delay: $e^{-\tau s}$: $|\cdot| = 1$, adds phase $-\omega\tau$. Max delay $\tau_{max} = \varphi_m/\omega_{gc}$.
Bode gain-phase: $\angle L(j\omega_{gc}) \approx 90^\circ\frac{d\log|L|}{d\log\omega}$; −20dB/dec→ -90°, $-40\to-180^\circ$.

16. Sensitivity / Gang of Four (Lec 11)

$$S = \frac{1}{1+PC}, \qquad T = \frac{PC}{1+PC}, \qquad S+T=1$$

Output: $\eta = Tr + PS\,d - Tn$; error $r - \eta = Sr - SPd + Tn$.
$\begin{pmatrix}\eta\\ e\\ u\end{pmatrix} = \begin{pmatrix}T & PS & -T\\ S & -PS & -S\\ CS & -T & -CS\end{pmatrix}\begin{pmatrix}r\\ d\\ n\end{pmatrix}$ (r=ref, d=dist, n=noise; set other 2= 0 to solve each TF)
Disturbance: $\eta = PS\,d$. $|S|{<}1$ attenuates; $|S|{=}0$ eliminates; $|S|{>}1$ worsens.
S measures sensitivity of CL to plant uncertainty.
Loop shaping — why: shape $L{=}PC$ to balance performance vs stability.
how: (i) high gain / integrator at low $\omega\to$ tracking + reject d ($S{\approx}0$); (ii) low gain + steep roll-off at high $\omega\to$ reject noise n ($T{\approx}0$); (iii) gentle slope (≈ −20dB/dec) near $\omega_{gc}\to$ keep phase margin.
Or pick desired L (e.g. $\frac{\alpha}{s}$, α sets ω_{gc}) then $C{=}L/P$ — avoid cancelling unstable poles/zeros.

17. Fundamental Limitations (Lec 11,12)

(1) $S+T=1$ — seesaw trade-off.
(2) Gain–phase relation — steep roll-off ⇒ phase lag, low PM.
(3) Interpolation constraints — RHP poles/zeros can't be hidden.
(4) Bode sensitivity integral (waterbed):

$$\int_0^\infty \log|S(j\omega)|\,d\omega = \pi\sum_k p_k$$

p_k=unstable open-loop poles; push $|S|$ down somewhere ⇒ up elsewhere.

19. Block Diagram Algebra

Series: G_1G_2. ***Parallel:*** $G_1\pm G_2$.
Neg. feedback: $\dfrac{G}{1+GH}$; unity: $\dfrac{G}{1+G}$.
Pos. feedback: $\dfrac{G}{1-GH}$.
$L(s) = C(s)G(s)$ (loop TF), $T(s) = \frac{L(s)}{1+L(s)}$, $G_{er} = \frac{1}{1+L(s)}$ (sens. fn).
Cascaded SS: $S_1\to S_2$ in series: get $G_i{=}C_i(sI-A_i)^{-1}B_i{+}D_i$ for each, then $G_{tot}{=}G_2G_1$ (multiply) — easier than building one big A.

20. Quick Reference / Exam Tips

- Stable ⇔ all poles/eigenvalues in open LHP.
- Reachable: $\det[B\ AB]\neq0$; Observable: $\det[C; CA]\neq0$ (2nd order).
- PD design: pick ζ from M_p, ω_n from t_s, match denom.
- Nyquist through $(\alpha, 0)$: set $G(s) = \alpha$; root is purely imaginary $s{=}\pm j\omega$.
- Zero SS error tracking: use **integral** action (robust) or feedforward N (needs exact model).
- Real-world uncertainty: parameter errors, disturbances, sensor noise, unmodelled nonlinearity.

PID vs state-feedback: **PID**: model-free, mainly SISO, heuristic/robust, simple to tune, limited pole control. **State-FB**: needs accurate model + full state (or observer), handles MIMO, gives exact pole placement.
Useful: $\zeta{=}0.5\Rightarrow M_p{\approx}16\%$; $\zeta{=}0.707\Rightarrow M_p{\approx}4.3\%$; $\zeta{=}0.6\Rightarrow M_p{\approx}9.5\%$. $\sqrt{1-\zeta^2}$ at $\zeta{=}0.707$ is 0.707.

21. Math Toolbox

Quadratic: $as^2{+}bs{+}c{=}0 \Rightarrow s = \dfrac{-b\pm\sqrt{b^2-4c\,a}}{2a}$.
2×2 ***inverse:*** $\begin{pmatrix}a & b\\ c & d\end{pmatrix}^{-1} = \dfrac{1}{ad-bc}\begin{pmatrix}d & -b\\ -c & a\end{pmatrix}$.
2×2 ***eigenvalues:*** $\det(\lambda I - A) = \lambda^2 - \mathrm{tr}(A)\lambda + \det(A)$,

$$\lambda = \tfrac{\mathrm{tr}\pm\sqrt{\mathrm{tr}^2-4\det}}{2}.$$

Resolvent: $(sI-A)^{-1} = \dfrac{\mathrm{adj}(sI-A)}{\det(sI-A)}$.
Euler: $e^{j\theta} = \cos\theta + j\sin\theta$; $|a+jb| = \sqrt{a^2+b^2}$, $\angle = \arctan\frac{b}{a}$.
Complex divide: $\frac{1}{a+jb} = \frac{a-jb}{a^2+b^2}$.
dB: $20\log_{10}2{\approx}6$, $20\log_{10}10{=}20$, $20\log_{10}\frac12\approx-6$.
Partial fractions: $\frac{1}{(s+p_1)(s+p_2)} = \frac{A}{s+p_1} + \frac{B}{s+p_2}$, $A = \frac{1}{p_2-p_1}$.
Common integrals: $\int e^{-at}dt = -\frac1a e^{-at}$; $\int_0^\infty e^{-at}dt = \frac1a$ ($a{>}0$).
Sin/cos deriv: $\frac{d}{dt}\sin\omega t = \omega\cos\omega t$, $\frac{d}{dt}\cos\omega t = -\omega\sin\omega t$.
Sin/cos integ: $\int\sin\omega t\,dt = -\frac{1}{\omega}\cos\omega t$, $\int\cos\omega t\,dt = \frac1\omega\sin\omega t$.
Trig: $\sin\theta = \frac{e^{j\theta}-e^{-j\theta}}{2j}$, $\cos\theta = \frac{e^{j\theta}+e^{-j\theta}}{2}$.